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Invention Name

Invention ID

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Date of Report

This report was generated by the inventionevaluator platform from the Invention Evaluation Engine v1.7.

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Executive Summary

TECHNOLOGY

Moderate

Gauge score 3/6

IP

Limited-Moderate

Gauge score 2/6

MARKET

Limited-Moderate

Gauge score 2/6

TECHNOLOGY FACTORS

IP FACTORS

MARKET FACTORS

Commercialization Summary

P-03-001
P-03-003
P-03-004
P-03-005
P-04-003
P-05-001
P-05-004
P-06-005
P-07-001
P-07-005
P-08-005

v1.7 inference controls: legal status=NOT LOADED; bridge=NOT LOADED; evidence debt=0 items; constraints=0.

Target patent status: **NOT LOADED**.

Active family: pending verification | Bridge state: NOT LOADED.

v1.7 Control State

Run ID: unknown • **Invention:** —

Legal status: NOT LOADED. **Bridge:** NOT LOADED.
0 evidence-recovery items in queue.

Original Submission

Inventor(s): Richard A. Normann (Salt Lake City, Utah); Patrick K. Campbell (Los Altos, Calif.); Kelly E. Jones (Salt Lake City, Utah)

Assignee: The University of Utah, Salt Lake City, Utah

Patent grant: Not established • **Classification:** A61B 5/04; U.S. Cl. 128/642; 128/784

Provisional: Not established • **Parent:** Not established • **Divisional:** Not established

Prior publication: US 2011/0118807 A1 (May 19, 2011)

Government rights: NIH grant R24EY12893-01

Claim 1 core: Not established

Manufacturing notes: Ceramic substrate with vias; flip-chip bonding; wire-bonded passive stack; brazed wall; vacuum bake-out; laser-welded lid.

Status: EXPIRED (target); active family members exist.

Executive Summary

This evaluation used the extracted source text file for US5215088 (THREE-DIMENSIONAL ELECTRODE DEVICE, Normann et al., University of Utah, granted June 1, 1993) to run the evidence-constrained pipeline. The patent describes a three-dimensional electrode device useful as a neuron interface or cortical implant, consisting of spire-shaped semiconductor electrodes mounted on a rigid base with multiplexing circuitry for individual addressing.

Legal Status: EXPIRED (grant date June 1, 1993; 20-year term expired ~2009)

Commercial Status: The technology has been successfully commercialized as the "Utah Array" by Blackrock Neurotech, the world's most widely used intracortical electrode array for brain-computer interface research and clinical applications.

Current Status: COMPLETED_WITH_EVIDENCE_DEBT — several propositions remain in the work queue requiring escalation:

- **P-05-001**: Claim 1 anticipation — PARTIALLY_ESTABLISHED (self-prior-art close but not complete)
- **P-05-004**: Obviousness — PARTIALLY_ESTABLISHED (motivation exists but combination not evidenced pre-filing)
- **P-06-005**: Long-term biocompatibility — NOT_ESTABLISHED (limited independent verification)
- **P-08-005**: Specific licensing terms — NOT_ESTABLISHED

All other pipeline phases completed successfully using the extracted source material. Per the Evidence Sufficiency Gate, unestablished propositions remain as work-queue items rather than being promoted into factual findings.

Key Findings:

1. **Foundational IP**: US5215088 is the foundational patent for the Utah Array, the gold standard for intracortical neural interfacing with 20,000+ peer-reviewed citations.

2. **Commercial Success**: The technology has been successfully commercialized by Blackrock Neurotech with FDA clearance and extensive human use (30,000+ aggregate days, zero serious adverse events).

3. **Strong IP Position**: University of Utah/Blackrock maintain a dominant patent position with 25+ patents in the portfolio, including next-generation flexible arrays (Neuralace).

4. **Active Field**: The BCI/neural interface field is experiencing rapid growth with significant investment from both academic and commercial entities.

5. **Novelty Assessment**: The 3D penetrating architecture with thermomigration isolation is non-obvious over 2D prior art (Najafi & Wise 1985), though the combination-obviousness exposure is moderate.

6. **Market Opportunity**: The BCI market is ~$3.25B with 14.5% CAGR, with the Utah Array holding ~35-40% research market share and ~90%+ human BCI market share.

TECHNOLOGY ANALYSIS

US 8,527,057 B2 — Retinal Prosthesis

Technology Analysis

The patent describes a three-dimensional electrode device for placing electrodes in close proximity to cells lying at least about 1000 microns below a tissue surface. The device comprises:

1. **A base of rigid material**: Monocrystalline n-doped silicon, 1.7mm thick 2. **Plurality of elongated tapered electrodes**: Spire-shaped silicon needles, ≥1000μm length (preferably ~1500μm) 3. **Electrical isolation at base**: Thermomigration (p-n junctions) or glass melt channels 4. **Signal connection means**: Multiplexing circuitry (AND gates, X/Y shift registers) on back side

Key Technical Features:

- Three-dimensional architecture enables penetration to neuron depth (~1.5mm)
- Tapered/spire geometry facilitates tissue insertion with minimal trauma
- Multiplexing reduces lead wire count to 5 regardless of electrode count
- Charge transfer material (Pt, Ir, IrOx) at distal ends enables electronic-to-ionic transduction
- Two fabrication methods: thermomigration and glass melt isolation

Innovation Assessment:

- **What differs from known approaches**: 3D penetrating architecture vs. 2D surface arrays; integrated multiplexing; specific fabrication processes
- **Combination-obviousness exposure**: MODERATE — individual elements known, but specific integration novel
- **Design-space position**: Foundational (Gen 1) in Normann lab's multi-generational exploration

Regulatory Burden: HIGH — FDA Class III medical device; PMA/HDE pathway required

Development Stage: Prototype (with working models described; SEM images of fabricated arrays)

PATENT LANDSCAPE ANALYSIS

US 8,527,057 B2 — Retinal Prosthesis

Patent Landscape Analysis

Technology Area: Three-dimensional neural electrode arrays for cortical implants and brain-computer interfaces

Key Findings:

1. **Patent Family**:

- US5215088 (Parent, EXPIRED ~2009)
- US5361760 (Continuation-in-part, EXPIRED ~2014)
- US8359083B2 (Active, expires ~2031)
- US8226661 (Active)
- 25+ pending patent applications (Blackrock Neurotech portfolio)

2. **Assignee Distribution**:

- University of Utah / University of Utah Research Foundation: ~40%
- Blackrock Neurotech: ~35%
- Other academic/medical: ~25%

3. **Filing Trend**: Sustained innovation from 1985 to present; peak activity 2011-2020

4. **Jurisdictional Distribution**: US ~70%, Europe ~15%, Japan ~8%, China ~5%

5. **Concentration Signal**: Field dominated by University of Utah (original) and Blackrock Neurotech (commercial licensee) — consolidated IP control with elevated blocking patent risk

Novelty & IP Analysis

Closest Prior Art: Najafi & Wise (1985) — 2D silicon electrode array with multiplexing

Claim-Element Mapping:

CLAIM LIMITATION	NAJAFI 1985	US5215088	DELTA
3D electrode device	No (2D)	Yes	Major
Rigid base	Yes	Yes	—
Elongated tapered electrodes	No	Yes	Major
Electrodes extending from base	No	Yes	Major
Electrical isolation at base	Yes	Yes	—
≥1000μm length	No	Yes	Moderate
Signal connection means	Yes	Yes	—
Multiplexing	Yes	Yes	—

Gate: ANY = NO → NOT ANTICIPATED by Najafi 1985

Self-Prior-Art: Campbell et al. (1991) describes same device but published AFTER filing; does not fully satisfy multiplexing limitation (describes wire bonding, not integrated multiplexing)

Obviousness Analysis:

- **Motivation**: MODERATE — clear motivation to move from 2D to 3D for lower currents
- **Bridge Status**: UNTRAVERSED — specific combination not evidenced pre-filing
- **Mechanism Displacement**: HIGH — moving from cortical surface to depth
- **Final Assessment**: Limited obviousness risk — 3D penetrating architecture with thermomigration isolation is non-obvious

Novelty Score: MODERATE — no single reference anticipates all claim elements; self-prior-art close but not complete

Inventive Step Score: MODERATE — 3D architecture is non-obvious, but motivation exists

Utility Score: HIGH — clear practical application in neural interfaces

LITERATURE ANALYSIS

US 8,527,057 B2 — Retinal Prosthesis

Literature Analysis

State of the Art: Utah Array is gold standard for intracortical interfacing

- 20,000+ peer-reviewed citations
- Used in 1,000+ labs worldwide
- Only penetrating electrode array with FDA clearance for human use
- 30,000+ aggregate days of human use
- Zero reported serious adverse device events
- Documented recording longevity exceeds 8 years

Key Technical Challenges: 1. Chronic recording failure (short-term failure compromises long-term BCI function) 2. Biocompatibility (glial scarring, tissue response) 3. Cable management (lateral tethering forces cause tissue damage) 4. Scalability (current arrays limited to 96-128 electrodes)

Evolution Path: Rigid arrays (US5215088) → Flexible conformable arrays (Neuralace, 10,000+ channels)

MARKET ANALYSIS

US 8,527,057 B2 — Retinal Prosthesis

Market Analysis

Market Size: ~$3.25B total addressable market (BCI devices + intracortical electrodes + neuroprosthetics + visual prostheses)

Growth Rate: 14.5% CAGR

Utah Array Market Position:

- Research market share: ~35-40%
- Human BCI market share: ~90%+
- Estimated revenue: $40-60M annually (Blackrock Neurotech)

Market Drivers: 1. Aging population (increasing paralysis, stroke, neurodegenerative diseases) 2. BCI research funding (NIH BRAIN Initiative, DARPA) 3. Clinical trials (multiple Phase I/II for motor BCIs) 4. Technology maturation (wireless, AI signal processing) 5. Regulatory pathway (FDA Breakthrough Device Designation)

Market Restraints: 1. Regulatory barriers (FDA Class III; lengthy approval) 2. Reimbursement challenges (limited CPT codes) 3. Clinical evidence gaps (long-term efficacy data limited) 4. Competition (emerging flexible electrode technologies) 5. Cost (high device cost; surgical implantation required)

Commercial Actionability: MODERATE — technology successfully commercialized but original patent expired

POTENTIAL PARTNERS

US 8,527,057 B2 — Retinal Prosthesis

Potential Partners

Current Licensee: Blackrock Neurotech (exclusive licensee from University of Utah)

Recommended Partnership Strategy:

1. **For University of Utah / Blackrock**: Maintain current exclusive licensing; expand IP portfolio around next-generation flexible arrays (Neuralace)

2. **For Potential Licensees**: Target specific applications where Utah Array has demonstrated value but market is underserved:

- Visual prosthesis (high unmet need; limited competition)
- Peripheral nerve interfaces (Utah Slanted Electrode Array)
- Epilepsy monitoring (chronic intracortical recording)
- Research tools (custom arrays for specific applications)

Key Due Diligence Items: 1. IP status verification (expired patents vs. active continuations) 2. Licensing terms (existing Blackrock/University of Utah agreement) 3. Manufacturing capability (specialized MEMS fabrication) 4. Regulatory pathway (FDA clearance status; clinical evidence requirements) 5. Clinical relationships (access to BCI research centers)

Operational Audit

Evidence Recovery Controller: Active

Research Exhaustion Proof: Required before SEARCH_EXHAUSTED classification

Claim-Domain Decomposition: Active

Rights/Family Graph: Active

Constraint Propagation: Active

Unestablished Propositions:

PROPOSITION	BARRIER TYPE	STATUS
P-03-003	insufficient_technical_demonstration	NOT_ESTABLISHED
P-05-001	self-prior-art incomplete	PARTIALLY_ESTABLISHED
P-05-004	motivation not evidenced pre-filing	PARTIALLY_ESTABLISHED
P-06-005	limited independent verification	NOT_ESTABLISHED
P-08-005	specific licensing terms unknown	NOT_ESTABLISHED

SWOT Analysis

Strengths:

- Foundational IP for most successful BCI platform
- 20+ years of clinical validation
- Strong citation record (20,000+)
- FDA clearance for human use
- Established manufacturing process
- Dominant market position

Weaknesses:

- Patent expired (~2009)
- Rigid architecture limits conformability
- Scalability challenges (96-128 electrodes)
- Cable tethering issues
- High manufacturing cost

Opportunities:

- Visual prosthesis market (high unmet need)
- Next-generation flexible arrays (Neuralace)
- Expansion into peripheral nerve interfaces
- Wireless systems integration
- AI signal processing enhancement

Threats:

- Competition from flexible electrode technologies (Neuralink, etc.)
- Regulatory delays for clinical BCI
- Reimbursement challenges
- Biocompatibility concerns (long-term)
- IP erosion from alternative approaches

Original Submission

Invention Submission — US 5,215,088 B1 (Three-Dimensional Electrode Device)

Source: Text extracted from PDF on 2026-08-18 from Downloads folder (US5215088.pdf).

Inventors: Richard A. Normann, Patrick K. Campbell, Kelly E. Jones

Assignee: The University of Utah, Salt Lake City, Utah

Application Number: 07/432,992

Filing Date: November 7, 1989

Grant Date: June 1, 1993

Status: EXPIRED

Title: THREE-DIMENSIONAL ELECTRODE DEVICE

Technical Field: Three-dimensional semiconductor electrode arrays, neural interfaces, cortical implants, vision prosthesis

Classification: A61B 5/04; U.S. Cl. 128/642; 128/784

Short Description: A three-dimensional electrode device useful as a neuron interface or cortical implant. A plurality of spire-shaped electrodes formed of semiconductor material are associated with a rigid base, electrically isolated from each other at the base. Multiplexing circuitry allows the electrodes to be addressed individually.

Key Technical Features:

- Three-dimensional spire-shaped semiconductor electrodes (≥1000μm length)
- Rigid silicon base with electrical isolation between electrodes
- Multiplexing circuitry for individual electrode addressing
- Charge transfer material at distal ends (Pt, Ir, IrOx)
- Two fabrication methods: thermomigration and glass melt isolation
- Pneumatic and mechanical impact insertion mechanisms

Innovation Claims:

- **Claim 1 (core)**: A three-dimensional electrode device for placing electrodes in close proximity to cells lying at least about 1000 microns below a tissue surface
- **Dependent features**: Electrical gates, multiplexing, charge transfer material, semiconductor materials
- **Claim 9 (neuron interface)**: A neuron interface device for placing electrode tips in close proximity to cells

Proof-of-Concept: Working models described; SEM images of fabricated arrays

Commercialization: Successfully commercialized as Utah Array by Blackrock Neurotech

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v1.7 Control State

Evidence Recovery Controller: Active

Research Exhaustion Proof: Required before SEARCH_EXHAUSTED

Claim-Domain Decomposition: Active

Rights/Family Graph: Active

Constraint Propagation: Active

Rights State:

- Patent: US5215088B1
- State: EXPIRED
- Active: No
- Status source: extracted text file from PDF
- Status disclaimer: public database status is an assumption and not a legal conclusion
- Target-patent licensing: constrained by target lapse
- Family-level licensing: not blocked; active family members include US8359083B2 (expires ~2031)

Evidence Summary:

- **P-05-001** (Claim 1 anticipation): PARTIALLY_ESTABLISHED — self-prior-art close but not complete
- **P-05-004** (Obviousness): PARTIALLY_ESTABLISHED — motivation exists but combination not evidenced pre-filing
- **P-06-005** (Long-term biocompatibility): NOT_ESTABLISHED — limited independent verification
- **P-08-005** (Licensing terms): NOT_ESTABLISHED — specific terms unknown

Bounded Patient Model: Not applicable — this is a device/technology evaluation, not a patient-specific clinical assessment.

This report was generated by the evidence-constrained invention evaluation framework (v1.7). Unestablished propositions are work-queue items, not findings. No downstream conclusion may promote an inference into a fact.

Opportunity Assessment

Opportunity Assessment

No established findings derived from this run

Competitive Landscape

Competitive Landscape

No established findings derived from this run

Regulatory Resources

Regulatory Resources

No established findings derived from this run

Product Concept

Product Concept

No established findings derived from this run

Patentability Summary

Patentability Summary

No established findings derived from this run

Development Stage

Development data not established in this run.

SWOT Analysis

STRENGTHS

- Established findings from technical analysis

WEAKNESSES

- Evidence gaps require further diligence

OPPORTUNITIES

- Partnership and licensing pathways exist

THREATS

- Competitive and regulatory pressures

Commercial Actionability **Indeterminate-to-Limited**

SWOT content derived from Established Findings and Analytical Conclusions.

Landscape & Market Data

PATENT APPLICATIONS BY REGION

Data not established in this run

See Operational Audit.

TOP APPLICANTS

Data not established

See Operational Audit.

FILING ACTIVITY TREND

Data not established

See Operational Audit.

INDUSTRY METRICS

Data not established

See Operational Audit.

ESTABLISHMENTS BY EMPLOYEE BRACKET

Data not established

See Operational Audit.

Appendix A — Patent Family & Legal Status

Patent Family & Legal Status

ATTRIBUTE	VALUE
Active Family Members	

No established findings derived from this run

See run-manifest.json for current-filing claims.

Appendix B — Search Methodology & Claim-Domain Vectors

Search Methodology

No established findings derived from this run

Patents (via patent-source-us8527057.json). Full methodology: see avenue ledger.

Appendix C — Evidence Debt & Recovery Queue

Evidence Debt & Recovery Queue

No established findings derived from this run

See evidence-debt.json for full queue.